

WINNER ENGINEERING LEARNING HUB SERIES

FUNDAMENTAL TOPIC

PIPING ACCESSORIES

& PROVISION

CONSIDERATION



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service

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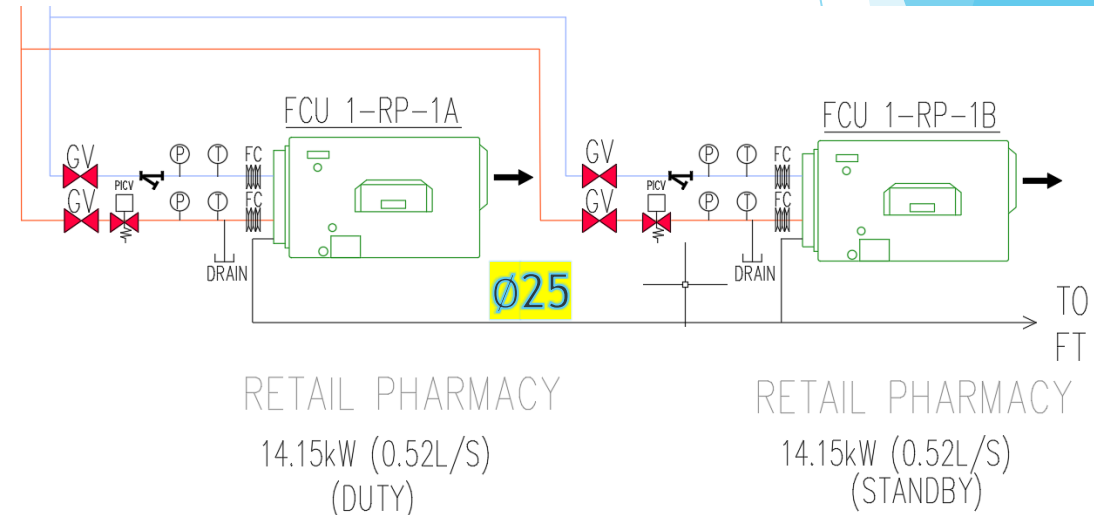
condensate drain pipe

All drain pipes, gradient make sure to have 1:100 or 1% for smooth flow of water.

condensate drain pipe sizing

TABLE OF CONDENSATE DRAIN PIPE

COOLING CAP OF FCU	PIPE SIZES
BELOW 15 kW	DIA 25
15.1kW TO 35kW	DIA 32
35.1kW TO 175kW	DIA 38
ABOVE 175kW	DIA 50



Condensate drain pipe can be installed in two ways

1. Exposed
2. concealed

1.EXPOSED DRAIN PIPE:



- Drain pipe hanged above the ceiling or run along the wall exposed.

DRAIN PIPE

2.CONCEALED DRAIN PIPE:



FLOOR TRAP (OTHERS)

- concealed drain pipes installed inside the concrete wall ,floor and brick wall.

DRAIN PIPE

CONCRETE FLOOR

Drain Pipe Insulation

As per the insulation of the drain pipe , 3/8”(10mm) thick closed- cell insulations are usually sufficient to prevent condensation. if our drain pipe is exposed to high temperatures (more the 30 °C), we should use 1/2” (13mm) thickness instead.



Drain pipe Accessories & Provision:

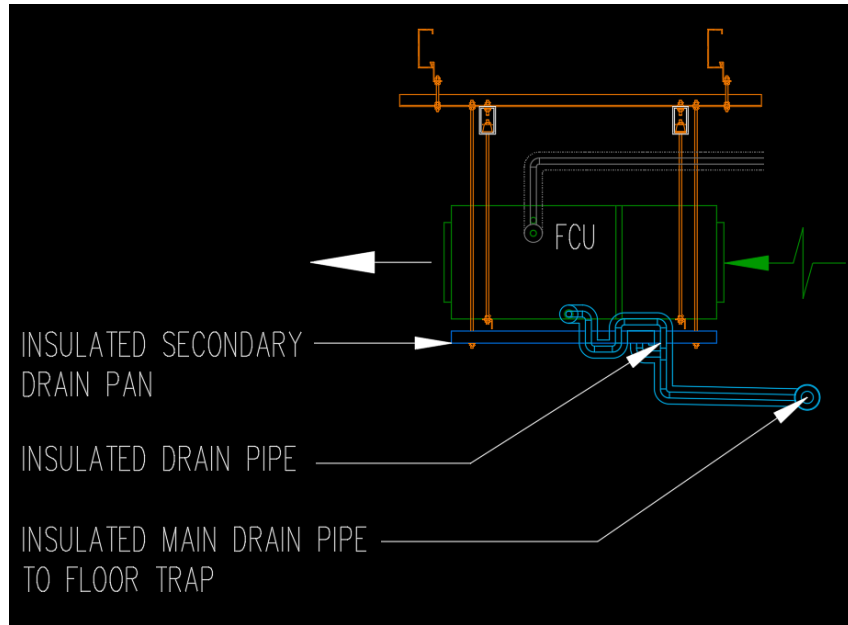
U trap:



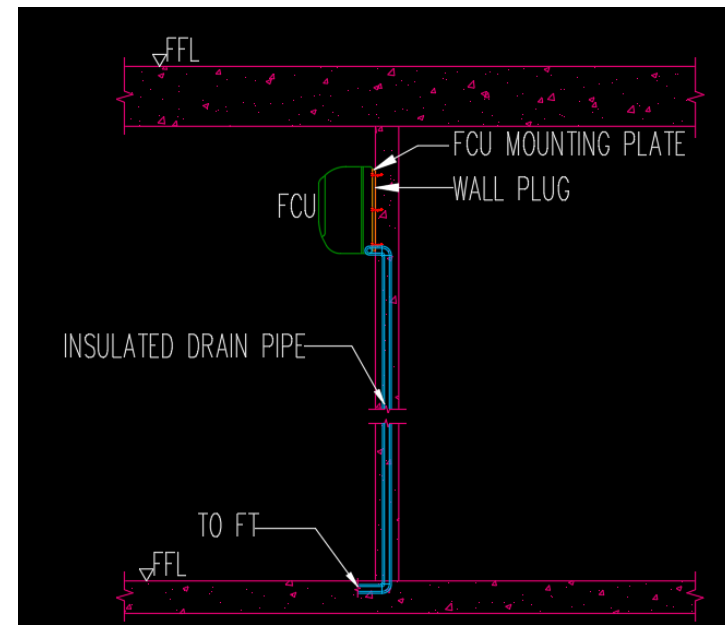
Purpose of installing U- Trap:

1. Main purpose is stop sewer gases/bad odour/ bad smell back to the unit and space
2. Prevent the water back flow
3. Prevent the penetrations of any insects.

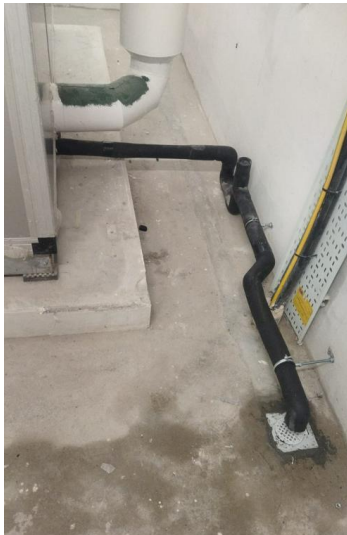
DUCTED FCU DRAIN CONNECTION:



WALL MOUNTED FCU DRAIN CONNECTION:

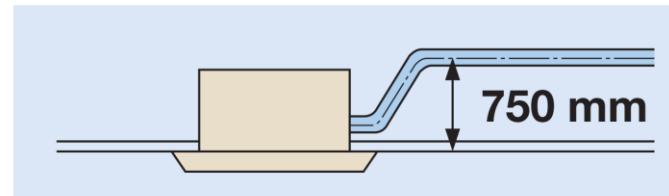


FLOOR MOUNT AHU DRAIN CONNECTION:



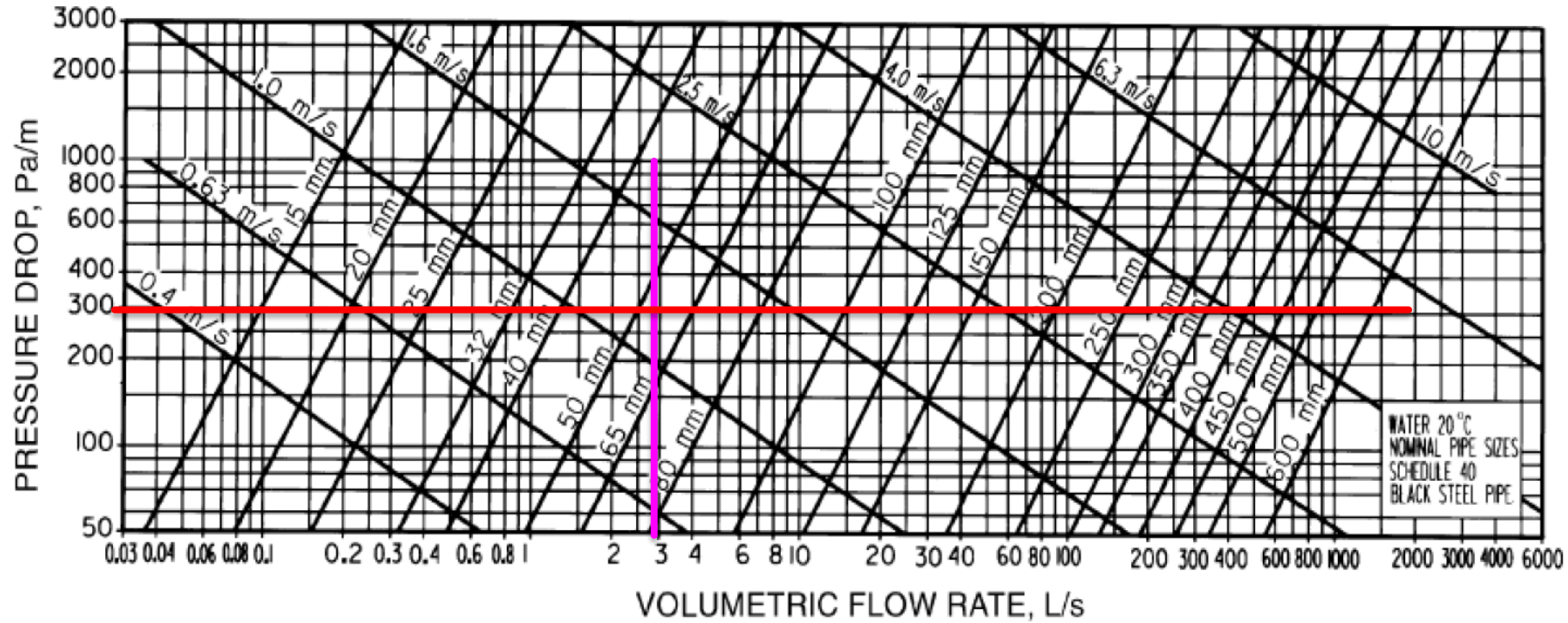
CEILING CASSETTE FCU DRAIN CONNECTION:

- Drain pump is equipped as standard accessory with 750 mm lift.



CHILLED WATER PIPE SIZEING:

Table 1: Friction Loss for Water in Commercial Steel Pipe (Scheduled 40)



SCHEDULE OF CHILLED WATER AIR HANDLING UNITS

Specified Capacity		Offered Capacity		On-Coil		Off-Coil		Chilled Water		Water Flow Rate	Water Pipe Size	Water Pressure Drop	Supply Air			Coil Face Velocity
Total	Sensible	Total	Sensible	DB	WB	DB	WB	In	Out				Specified	Offered		
(kw)	(kw)	(kw)	(kw)	(°C)	(°C)	(°C)	(°C)	(°C)	(°C)	(l/s)	(Ø mm)	(kPa)	(cmh)	(cmh)	(l/s)	(m/s)
73	49	79.78	56.33	23.10	17.20	10.30	10.28	8.0	15.0	2.72	65	46.73	12970	12970	3602.7778	1.4
88	63	95.42	72	23.30	16.90	10.29	10.27	8.0	15.0	3.25	65	32.05	16300	16300	4527.7778	1.37
77	41	84.24	45.59	22.10	18.40	10.39	10.39	8.0	15.0	2.87	65	49.34	11470	11470	3186.1111	1.3
97	52	107.33	57.69	22.20	18.50	10.34	10.34	8.0	15.0		65	37.95	14340	14340	3983.3333	1.26
73	39	80.51	43.18	22.00	18.40	10.36	10.36	8.0	15.0		65	45.52	10930	10930	3036.1111	1.24
105	57	115.8	63.21	22.00	18.30	10.49	10.49	8.0	15.0	3.94	65	29.65	16190	16190	4497.2222	1.28

CHILLED WATER PIPE INSULATION SIZEING:

ONSITE INSTALLTION

Available Insulation Thickness

GLENSUL[®] insulation is available in various thicknesses. The following shows the available range:

	Series 25		Series 40		Series 50		Series 65		Series 75	
Pipe Size	Casing Size	Approximate Insulation thickness	Casing Size	Approximate Insulation thickness	Casing Size	Approximate Insulation thickness	Casing Size	Approximate Insulation thickness	Casing Size	Approximate Insulation thickness
(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)	(mm)
15	80	29	100	39	125	52	150	64	170	74
20	80	27	100	37	125	49	150	62	170	72
25	90	28	115	41	140	53	160	63	190	78
32	90	24	115	36	140	49	170	64	190	74
40	100	26	125	38	140	46	170	61	200	76
50	115	27	140	40	160	50	190	65	215	77
65	125	25	150	37	170	47	200	62	224	74
80	140	26	170	41	190	51	215	63	240	76
100	170	28	190	38	215	50	240	63	265	75
125	190	25	224	42	240	50	265	63	295	78
150	215	25	240	37	265	50	295	65	320	77
200			295	38	320	50	350	65	375	78
250			350	39	375	51	400	64	425	76
300			400	38	425	51	450	63	475	76
350			425	35	450	47	475	60	510	77
400			475	34	510	52	530	62	560	77
450			530	37	560	52	590	67	610	77
500			590	41	610	51	635	64	660	76
600					710	50	735	63	760	75
700					800	45	850	70	900	95
800					900	44	950	69	1000	94
900					1000	43	1050	68	1100	93
1000					1100	42	1150	67	1200	92

■ Recommended insulation thickness under normal operating condition.



Chilled water pipe Accessories & Provision:



- ❑ Flexible connector
- ❑ Strainer
- ❑ Motorized valve
- ❑ Gate valve
- ❑ Balancing valve
- ❑ Temperature Gauge
- ❑ Drain cock
- ❑ Air vent

Chilled water pipe Accessories & Provision:

2 way
Motorized Valve



Strainer



Pressure Gauge



Flexible Connector



Balancing Valve



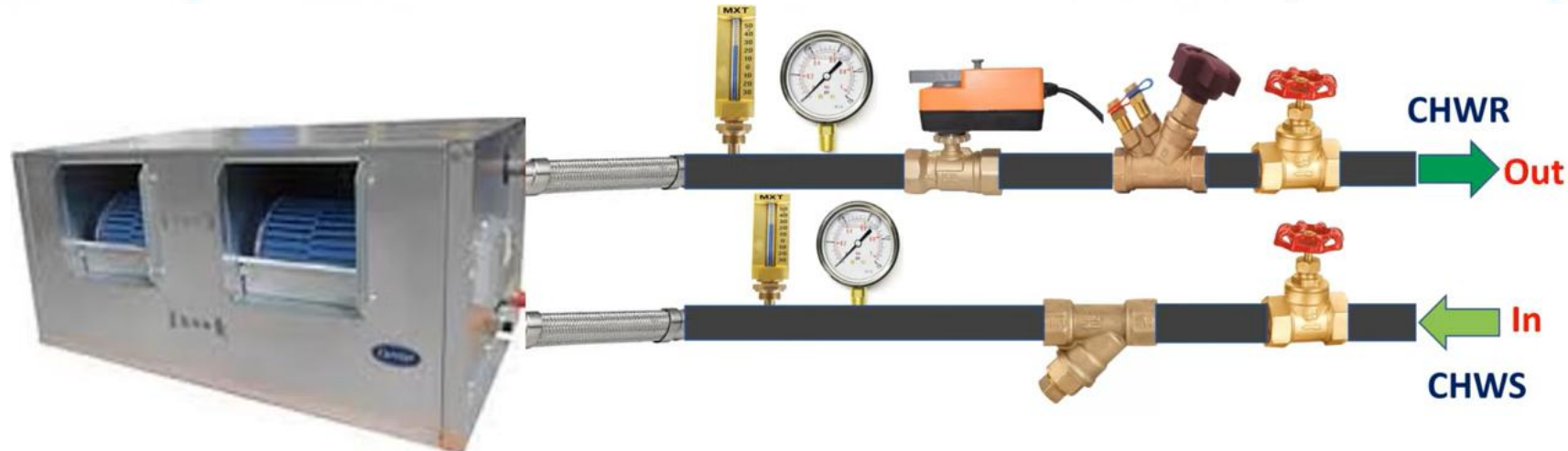
Gate Valve



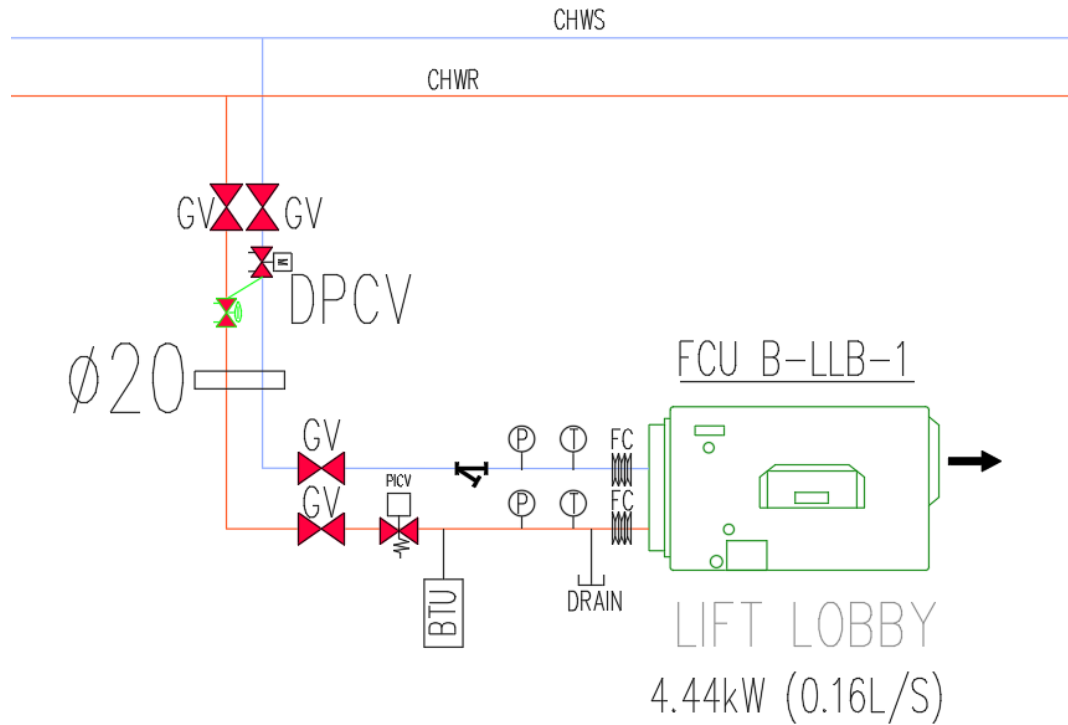
Thermometer



Chilled water Pipe with Jacketing



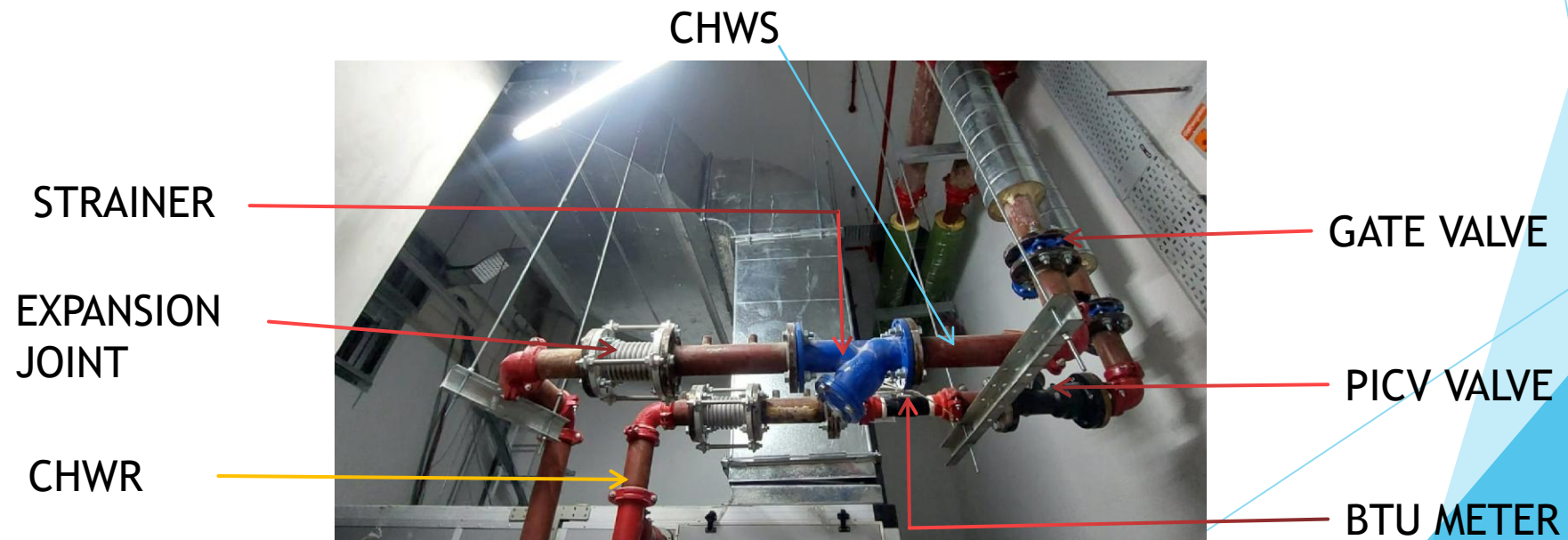
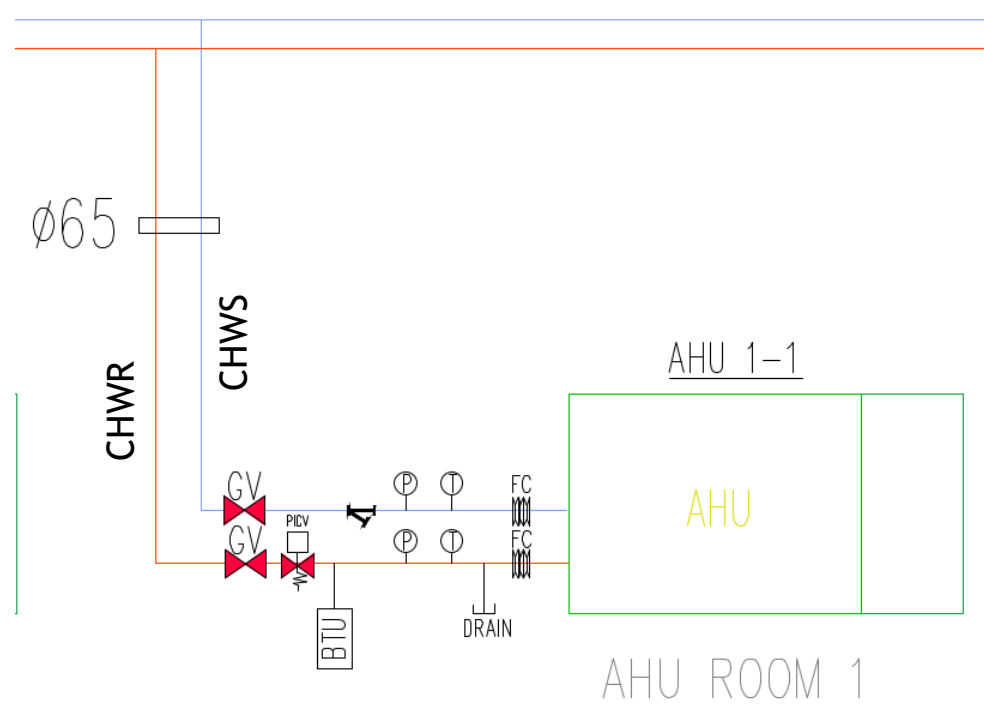
Chilled water pipe Accessories & Provision:



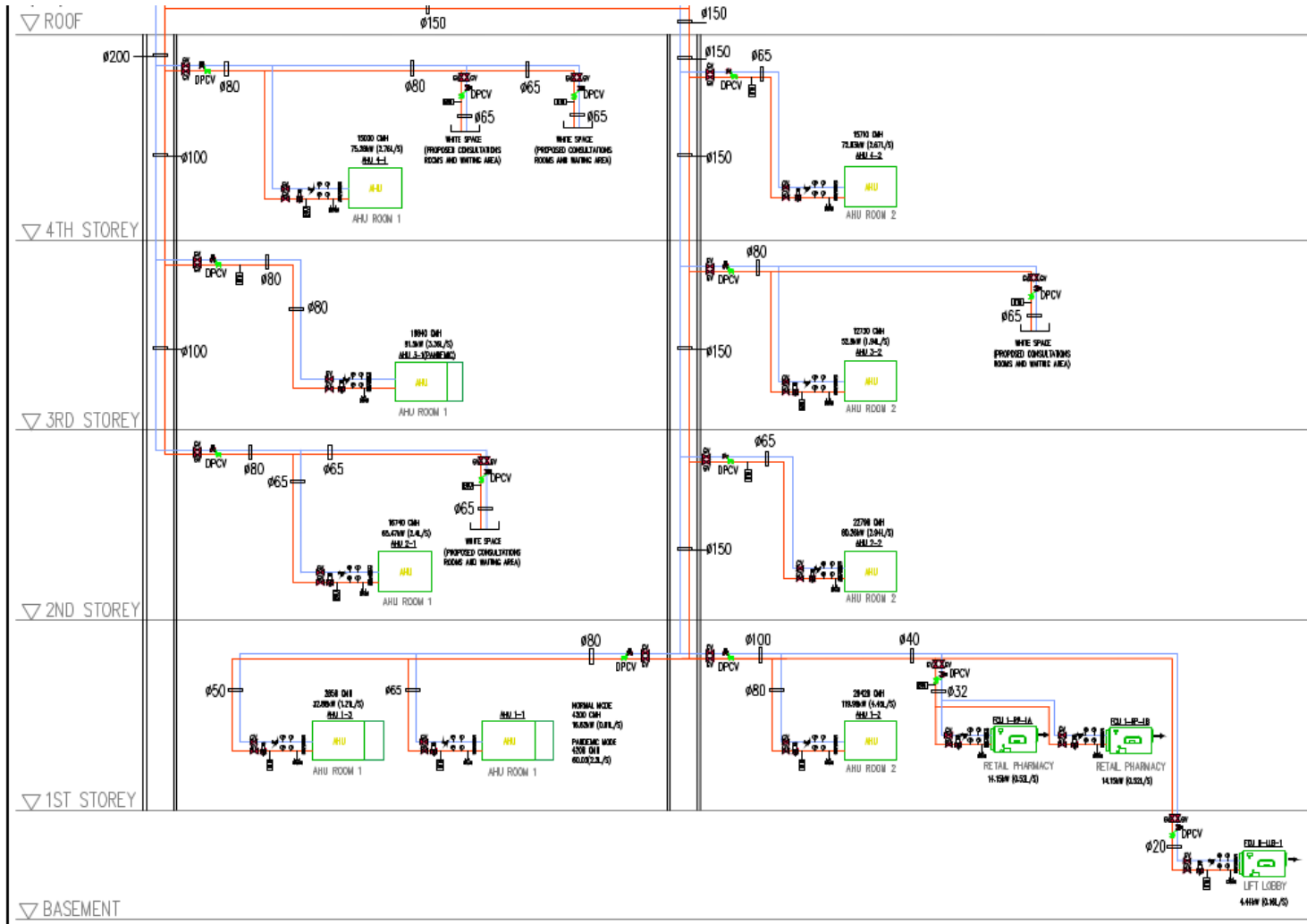
PICV VALVE:

Pressure independent control valves are an automatic temperature control valve and an automatic flow regulating valve packaged in one valve body.

TYPICAL AHU CONNECTION:

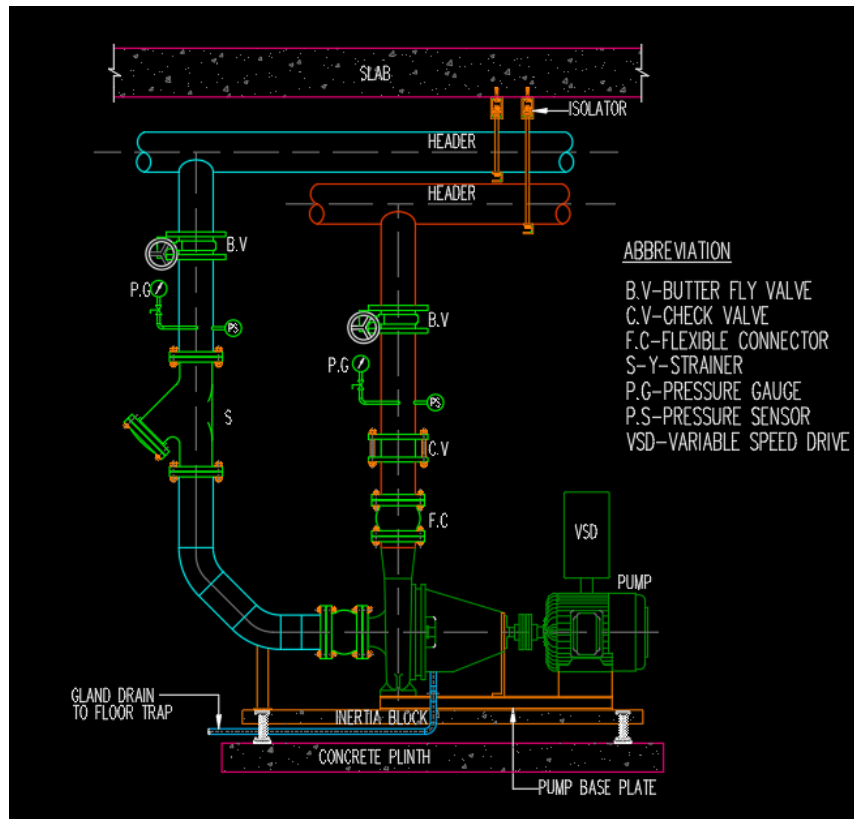


SCHEMATIC:



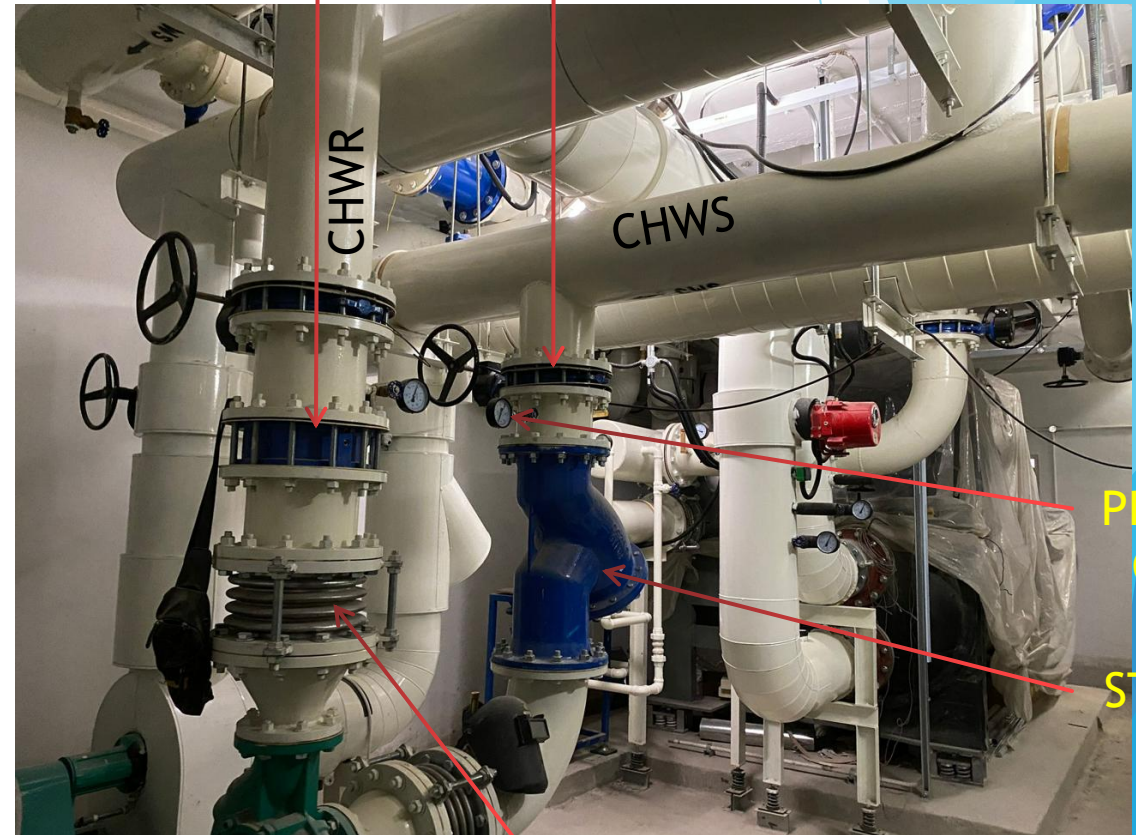
	GATE VALVE
	PRESSURE INDEPENDENT CONTROL VALVE
	DIFFERENTIAL PRESSURE CONTROL VALVE
	ENERGY METER
	PRESSURE SENSOR
	THERMOMETER GAUGE
	FLEXIBLE CONNECTOR
	STRAINER
	MOTORIZED GATE VALVE
	FLOW METER
	CHECK VALVE
	BUTTERFLY VALVE
	MOTORIZED BUTTERFLY VALVE
	REDUCER

TYPICAL PUMP CONNECTION:



CHECK VALVE

BUTTERFLY VALVE

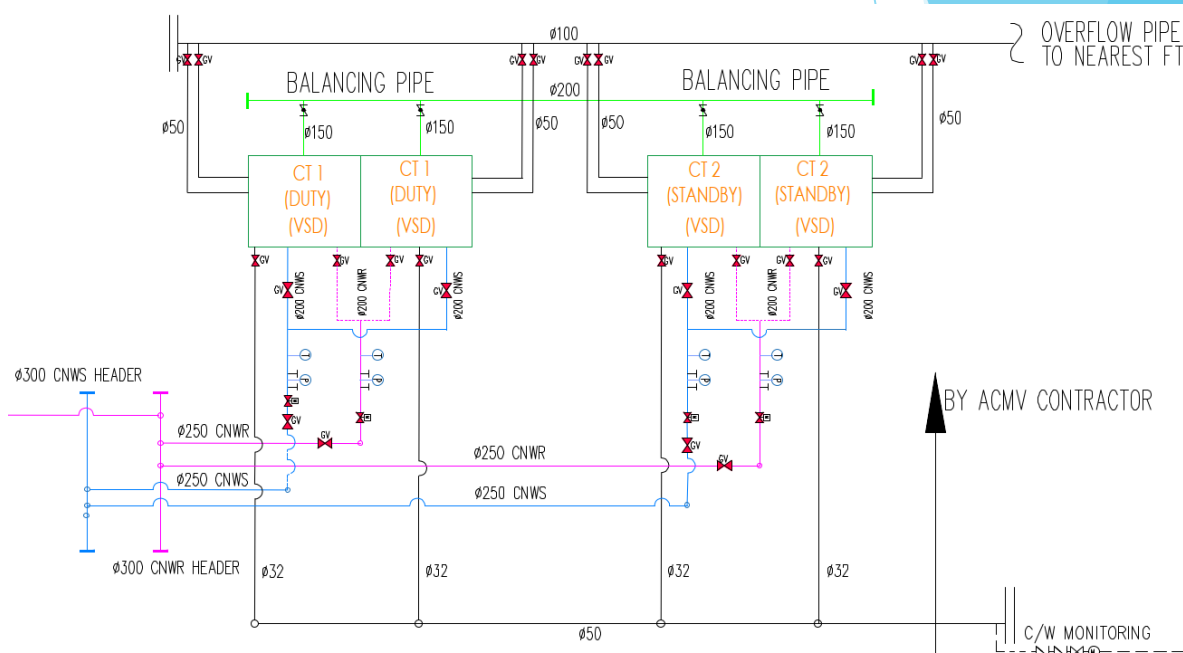
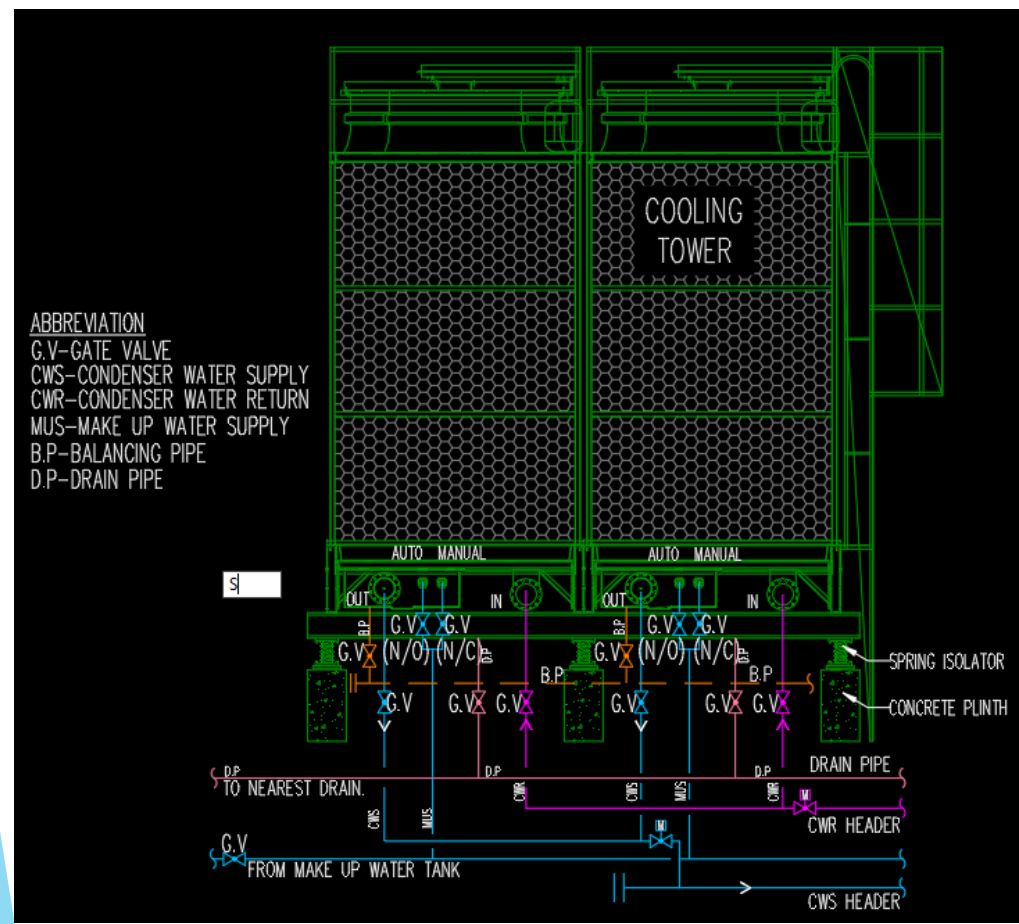


PRESSURE
GAUGE

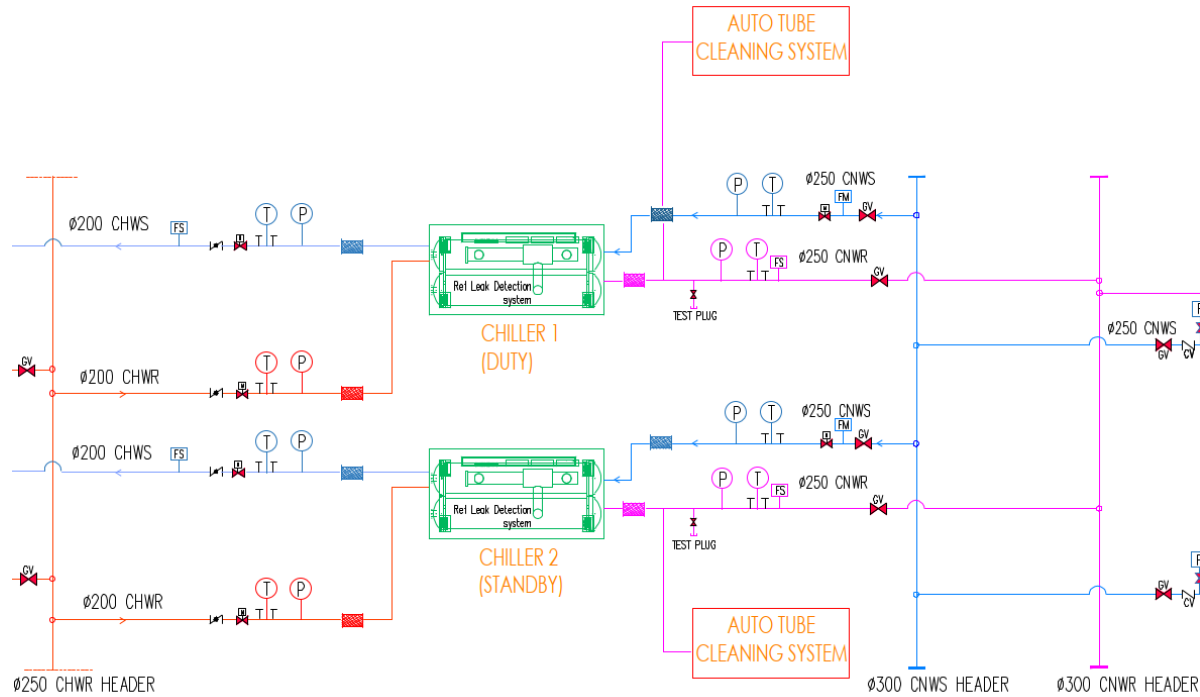
STRAINER

EXPANSION
JOINT

TYPICAL COOLING TOWER CONNECTION:



TYPICAL CHILLER CONNECTION:



BUTTERFLY VALVE

MOTORIZED VALVE

